

Chapter 5

p. 249 17LT Replace "state or a metastable state" with the following text:
"state, called a metastable state,"

p. 281 14LT insert the following footnote after "VHDL **process**":

A level-sensitive process depends on only the values of the signals (e.g., combinational logic). An edge sensitive process depends on signal transitions. For example, a flip-flop considers its inputs only when the clock has a transition, and it ignores the inputs otherwise.

p. 283 3LB Delete "**end if**;"

p. 283 16LB Replace "**end**" with "**end D_FF_vhdl**;"

p. 283 13LB Replace "Clk_b" with "Clk'event".

p. 283 13LB Replace "Q <= D;" with "Q <= D; **end if; end process**;"

p. 285 2LT Change: "assigned with by" to: "assigned by".

p. 285 9LB Replace "**out** Std_Logic" with "**buffer** Std_Logic".

p. 285 12LB Replace 12LB with "DFF M0 (Q, JK, Clk, rst);

p. 286 3LT Replace "Q: **out**" with "Q: **buffer**".

p. 286 2LB Replace "**out** Std_Logic" with "**buffer** Std_Logic".

p. 286 5LB Replace "**encase**" with "**endcase**".

p. 286 10LT Replace "//" with "--".

p. 287 4LT After 4LT insert a new line having the following text:

" **if** (rst = '1') **then** Q <= '0';"

p. 287 Replace "**end Behavioral_Case**" with "**end Behavioral_Case_vhdl**".

p. 287 5LT Replace "**case** (J & K)" with "**else case** (J & K)".

p. 287 6LT Replace '00' with "00"

p. 287 7LT Replace '01' with "01"

p. 287 8LT Replace '10' with "10"

p. 287 9LT Replace '11' with "11"

p. 291 18LT Replace "above" with "below".

p. 291 13LB Replace "state_type" with "state_type **is**"

p. 291 4LB, 3LB Replace "next_state = S1" with "next_state <= S1".

Replace "next_state = S0" with "next_state <= S0".

p. 291 2LB, 1LB Replace "next_state = S3; **else** next_state <= S0"
with "next_state <= S3; **else** next_state <= S0"

p. 291 9LB Replace "**else if** " with "**elsif**".

p. 292 1LT Replace "next_state = S0; **else** next_state = S2"
with "next_state <= S0; **else** next_state <= S2"

p. 292 3LT Replace "next_state = S2; **else** next_state <= S0" with
"next_state <= S2; **else** next_state <= S0".

p. 292 10LT Align S0 with S1 in 11LT

p. 292 10, 11, 12, 13LT Change "y_out =" to "y_out <=".

- p. 292 13LT Change "not x_in;" to "**not** x_in;"
- p. 294 6LB Replace "Clk" with "clock".
- p. 294 4LB Replace "rst" with "reset".
- p. 294 4LB Replace 4LB with the following:

```
if reset'event and reset = '0' state <= S0          -- Asynchronous reset
```
- p. 294 3LB, 4LB Between 4LB and 3LB insert the following line:

```
    elsif clock'event and clock = '1'
```
- p. 294 11LB Remove bold - Replace "**out, bit_vector**" with "**out** bit_vector".
- p. 294 11LB Remove bold - Replace "**1 downto 0**" with "(1 **downto** 0)".
- p. 294 10LB Replace "Moore_Model_vhdl" with "Moore_Model_Fig_5_19_vhdl".
- p. 294 9LB Replace "Fig_5_19 **is**" with "Fig_5_19_vhdl **is**".
- p. 294 3LB Delete **else**
- p. 294 3LB Replace "(state)" with "(state) **is**".
- p. 294 6LB Replace "(Clk, reset)" with "(clock)".
- p. 294 1LB, 2LB Replace 1LB and 2LB with the following text:

```

when S0 => if x_in = '0' then state <= S1; else state <= S0; end if;
when S1 => if x_in = '1' then state <= S2; else state <= S3; end if;

```

- p. 295 1LT, 2LT Replace 1LT and 2LT with the following text:

```

when S2 => if x_in = '0' then state <= S3; else state <= S2; end if;
when S3 => if x_in = '0' then state <= S0; else state <= S3; end if;
end case;
end if;

```

- p. 295 4LT Change "/" to "--".
- p. 298 5 LT Change "clock" to "Clk". Change "STD_LOGIC" to Std_logic". Change STD_logic" to "Std_logic".
- p. 298 12LT Change 12LT to "if rst_b = '0' then state <= S0;"
- p. 298 13LT Change "**else if**" to "**elsif**".
- p. 298 15LT Change "**if not** x_in" to "**if** x_in = '0'".
- p. 298 16LT Change "**if** x_in" to "**if** x_in = '1'".
- p. 298 17LT Change "**if not** x_in" to "**if** x_in = '0'".
- p. 298 18LT Change "**if not** x_in" to "**if** x_in = '0'".
- p. 298 10LB, 9LB, 8LB Change 10LB, 9LB, 8LB to the following

```

process (CLK, rst) begin
    if rst'event and reset = '0' then Q <= '0';
    elsif Clk'event and Clk = '1' then Q <= D; end if;

```
- p. 299 1LT Change "T_FF_vhdl" to "TFF_vhdl".
- p. 299 4LT Change "**out**" to "**buffer**".
- p. 299 14 LB Change "**else if**" to "**elsif**".
- p. 299 19LB Change "**out**" to "**buffer**".
- p. 315 In Problem 5.6 change "D flip-flops" to "D flip-flops,".
- p. 316 In Problem 5.10 change "(c)*" to "(c)".

p. 319 In Problem 5.31(b) replace "with = instead of <=" with

"with := instead of <=".

p. 322 12LB Replace "**else** state <= next_state;" with

"**elsif** clk'event **and** clk = '1' **then** state <= next_state;".

p. 323 19, 20, 21, 23LT In each line delete the following text: "**if** x_in = 1".

p. 323 4LB Replace 4LB with the following line:

elsif clk'event **and** clk = '1' **then** state <= next_state;

p. 323 1LB Remove bold font.